

0 2

Testosterone is a steroid hormone that belongs to a group of male sex hormones called androgens.

0 2 . 1

Steroid hormones are hydrophobic.

Explain why steroid hormones can rapidly enter a cell by passing through its cell-surface membrane.

[2 marks]

0 2 . 2

In the cytoplasm, testosterone binds to a specific androgen receptor (AR). An AR is a protein.

Suggest and explain why testosterone binds to a specific AR.

[2 marks]

Question 2 continues on the next page

0 2 3

The binding of testosterone to an AR changes the shape of the AR. This AR molecule now enters the nucleus and stimulates gene expression.

Suggest how the AR could stimulate gene expression.

[2 marks]

The gene that codes for the AR has a variable number of CAG repeats. Some studies have shown an association between the number of CAG repeats and the risk of developing prostate cancer.

Table 1 shows the results of a statistical test from one study.

Table 1

Number of CAG repeats in the <i>AR</i> gene	Probability (P) value
≤ 16	0.02
≤ 17	0.30
≤ 18	0.07
≤ 19	0.09
≥ 20	0.06

0 2 4

What can you conclude from the data in **Table 1**?

[3 marks]
